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United States Patent	12409853
Kind Code	B2
Date of Patent	September 09, 2025
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Unmanned vehicle management system and unmanned vehicle management method

Abstract

An unmanned vehicle management system includes: a traffic congestion prediction unit that predicts occurrence of a traffic congestion from travel situations of a plurality of unmanned vehicles traveling on a travel path at a work site; and a guidance command unit that, when occurrence of a traffic congestion is predicted, outputs a first guidance command so that an unmanned vehicle that causes the traffic congestion enters a side path branching from the travel path.

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Appl. No.:	18/271054
Filed (or PCT Filed):	December 28, 2021
PCT No.:	PCT/JP2021/048910
PCT Pub. No.:	WO2022/181051
PCT Pub. Date:	September 01, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240059314 A1	Feb. 22, 2024

Foreign Application Priority Data

JP2021-027917Feb. 24, 2021

Publication Classification

Int. Cl.: **B60W60/00** (20200101); **B60W30/14** (20060101); **B60W30/18** (20120101);
B60W40/04 (20060101); **B60W50/00** (20060101)

U.S. Cl.:

CPC **B60W60/001** (20200201); **B60W30/143** (20130101); **B60W30/18163** (20130101);
B60W40/04 (20130101); **B60W50/0097** (20130101); B60W2300/125 (20130101);
B60W2554/402 (20200201); B60W2554/406 (20200201); B60W2554/80 (20200201)

Field of Classification Search

CPC: B60W (30/143); B60W (30/18163); B60W (40/04); B60W (50/0097); B60W (60/00);
B60W (60/001); B60W (2300/125); B60W (2554/402); B60W (2554/406); B60W
(2554/80); G01C (21/34); G05D (1/6987); G05D (2105/05); G05D (2107/73); G05D
(2109/10); G08G (1/01); G08G (1/09)

USPC: 701/23

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Background/Summary

FIELD

(1) The present disclosure relates to an unmanned vehicle management system and an unmanned vehicle management method.

BACKGROUND

(2) As disclosed in Patent Literature 1, an unmanned vehicle operates at a work site in a wide area such as a mine.

CITATION LIST

Patent Literature

(3) Patent Literature 1: JP 2007-164280 A

SUMMARY

Technical Problem

(4) At the work site, an unmanned vehicle traveling at a high speed and an unmanned vehicle traveling at a low speed may operate in a mixed manner. A traffic congestion of the unmanned vehicles may occur at the work site due to the unmanned vehicle traveling at a low speed.

(5) An object of the present disclosure is to suppress occurrence of a traffic congestion at a work site.

Solution to Problem

(6) According to an aspect of the present invention, an unmanned vehicle management system comprises: a traffic congestion prediction unit that predicts occurrence of a traffic congestion from travel situations of a plurality of unmanned vehicles traveling on a travel path at a work site; and a guidance command unit that, when occurrence of a traffic congestion is predicted, outputs a first guidance command such that an unmanned vehicle that causes the traffic congestion enters a side path branching from the travel path.

Advantageous Effects of Invention

(7) According to the present disclosure, it is possible to suppress occurrence of a traffic congestion at a work site.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating a management system of a work site according to an embodiment.
- (2) FIG. 2 is a perspective view illustrating an unmanned haul vehicle according to the embodiment.
- (3) FIG. 3 is a perspective view illustrating an unmanned water-spraying vehicle according to the embodiment.
- (4) FIG. 4 is a schematic diagram illustrating a work site according to the embodiment.
- (5) FIG. 5 is a functional block diagram illustrating a management system of a work site according to the embodiment.
- (6) FIG. 6 is a diagram for explaining first course data according to the embodiment.
- (7) FIG. 7 is a diagram for explaining second course data according to the embodiment.
- (8) FIG. 8 is a diagram for explaining processing of a guidance command unit according to the embodiment.
- (9) FIG. 9 is a diagram for explaining processing of the guidance command unit according to the embodiment.
- (10) FIG. 10 is a diagram for explaining processing of the guidance command unit according to the embodiment.
- (11) FIG. 11 is a flowchart illustrating a management method of an unmanned vehicle according to the embodiment.

DESCRIPTION OF EMBODIMENTS

- (12) Hereinafter, embodiments according to the present disclosure will be described with reference to the drawings, but the present disclosure is not limited to the embodiments. The components of the embodiments described below can be appropriately combined. Furthermore, some components may not be used.
- (13) [Overview of Management System]
- (14) FIG. 1 is a schematic diagram illustrating an unmanned vehicle management system 1 according to an embodiment. The management system 1 manages an unmanned vehicle operating at a work site. The unmanned vehicle refers to a work vehicle that operates in an unmanned manner without depending on a driving operation by a driver. In the embodiment, the unmanned vehicle operating at the work site includes a first unmanned vehicle 10 and a second unmanned vehicle 20.
- (15) In the embodiment, the first unmanned vehicle 10 is an unmanned haul vehicle. The second unmanned vehicle 20 is an unmanned water-spraying vehicle. In the following description, the first unmanned vehicle 10 is appropriately referred to as an unmanned haul vehicle 10, and the second unmanned vehicle 20 is appropriately referred to as an unmanned water-spraying vehicle 20.
- (16) The unmanned haul vehicle 10 travels in an unmanned manner at the work site to haul a load. An unmanned dump truck is exemplified as the unmanned haul vehicle 10. An excavated object excavated at the work site is exemplified as a load to be hauled to the unmanned haul vehicle 10.
- (17) The unmanned water-spraying vehicle 20 travels at the work site in an unmanned manner and sprays water. An unmanned water-spraying truck is exemplified as the unmanned water-spraying vehicle 20. The unmanned water-spraying vehicle 20 sprays water to suppress diffusion of dust or sand at the work site.
- (18) The management system 1 includes a management device 2 and a communication system 3. The management device 2 is installed in a control facility 4 of the work site. There is an

administrator in the control facility **4**.

(19) The unmanned haul vehicle **10** includes a control device **11**. The unmanned water-spraying vehicle **20** includes a control device **21**. The management device **2**, the control device **11**, and the control device **21** wirelessly communicate with each other via the communication system **3**. A wireless communication device **3A** is connected to the management device **2**. A wireless communication device **3B** is connected to the control device **11**. A wireless communication device **3C** is connected to the control device **21**. The communication system **3** includes the wireless communication device **3A**, the wireless communication device **3B**, and the wireless communication device **3C**.

(20) [Unmanned Haul Vehicle]

(21) FIG. **2** is a perspective view illustrating the unmanned haul vehicle **10** according to the embodiment. As illustrated in FIGS. **1** and **2**, the unmanned haul vehicle **10** includes the wireless communication device **3B**, the control device **11**, a vehicle body **12**, a traveling device **13**, a dump body **14**, and a sensor system **15**.

(22) The vehicle body **12** includes a vehicle body frame. The vehicle body **12** is supported by the traveling device **13**. The vehicle body **12** supports the dump body **14**.

(23) The traveling device **13** generates a driving force for causing the unmanned haul vehicle **10** to travel. The traveling device **13** generates a braking force for decelerating or stopping the unmanned haul vehicle **10**. The traveling device **13** generates a steering force for turning the unmanned haul vehicle **10**. The traveling device **13** moves the unmanned haul vehicle **10** forward or backward. The traveling device **13** includes wheels **16**. A tire **17** is mounted on each of the wheels **16**. The wheel **16** includes a front wheel **16F** and a rear wheel **16R**. The tire **17** includes a front tire **17F** mounted on the front wheel **16F** and a rear tire **17R** mounted on the rear wheel **16R**. When the wheels **16** rotate in a state where the tires **17** are in contact with a road surface of the work site, the unmanned haul vehicle **10** travels through the work site.

(24) The dump body **14** is a member on which a load is loaded. At least a part of the dump body **14** is disposed above the vehicle body **12**.

(25) The sensor system **15** includes a position sensor **15A**, an orientation sensor **15B**, a speed sensor **15C**, and an obstacle sensor **15D**. The position sensor **15A** detects a position of the unmanned haul vehicle **10**. The position of the unmanned haul vehicle **10** is detected using a global navigation satellite system (GNSS). The position sensor **15A** includes a GNSS receiver and detects the position of the unmanned haul vehicle **10** in the global coordinate system. The orientation sensor **15B** detects an orientation of the unmanned haul vehicle **10**. A gyro sensor is exemplified as the orientation sensor **15B**. The speed sensor **15C** detects a traveling speed of the unmanned haul vehicle **10**. As the speed sensor **15C**, a pulse sensor that detects the rotation of the wheel **16** is exemplified. The obstacle sensor **15D** detects an obstacle around the unmanned haul vehicle **10**. The obstacle sensor **15D** detects an obstacle in a non-contact manner. Examples of the obstacle sensor **15D** include a laser sensor (light detection and ranging (LIDAR)) and a radar sensor (radio detection and ranging (RADAR)).

(26) [Unmanned Water-Spraying Vehicle]

(27) FIG. **3** is a perspective view illustrating the unmanned water-spraying vehicle **20** according to the embodiment. As illustrated in FIGS. **1** and **3**, the unmanned water-spraying vehicle **20** includes the wireless communication device **3C**, the control device **21**, a vehicle body **22**, a traveling device **23**, a tank **24**, a sensor system **25**, and a watering spray **28**.

(28) The vehicle body **22** includes a vehicle body frame. The vehicle body **22** is supported by the traveling device **23**. The vehicle body **22** supports the tank **24**.

(29) In the embodiment, a cab **29** is provided in the vehicle body **22**. The cab **29** is provided at a front portion of the vehicle body **22**. A driver can board the cab **29** and perform a driving operation of the unmanned water-spraying vehicle **20**. For example, when performing maintenance or inspection of the unmanned water-spraying vehicle **20**, the driver performs a driving operation of

the unmanned water-spraying vehicle **20**. In the embodiment, the unmanned water-spraying vehicle **20** operates in an unmanned manner at least when spraying water at the work site. Note that the cab **29** may not be provided in the unmanned water-spraying vehicle **20**.

(30) The traveling device **23** generates a driving force for traveling the unmanned water-spraying vehicle **20**. The traveling device **23** generates a braking force for decelerating or stopping the unmanned water-spraying vehicle **20**. The traveling device **23** generates a steering force for turning the unmanned water-spraying vehicle **20**. The traveling device **23** moves the unmanned water-spraying vehicle **20** forward or backward. The traveling device **23** includes wheels **26**. A tire **27** is mounted on each of the wheels **26**. The wheel **26** includes a front wheel **26F** and a rear wheel **26R**. The front wheel **26F** is a steering wheel, and the rear wheel **26R** is a driving wheel. Note that both the front wheel **26F** and the rear wheel **26R** may be steering wheels. Both the front wheel **26F** and the rear wheel **26R** may be driving wheels. The front wheel **26F** may be a drive wheel, and the rear wheel **26R** may be a steering wheel. The tire **27** includes a front tire **27F** mounted on the front wheel **26F** and a rear tire **27R** mounted on the rear wheel **26R**. When the wheels **26** rotate in a state where the tires **27** are in contact with the road surface of the work site, the unmanned water-spraying vehicle **20** travels through the work site.

(31) The tank **24** is a member that stores water for water-spraying. At least a part of the tank **24** is disposed above the vehicle body **22**.

(32) The sensor system **25** includes a position sensor **25A**, an orientation sensor **25B**, a speed sensor **25C**, and an obstacle sensor **25D**. The position sensor **25A** detects a position of the unmanned water-spraying vehicle **20**. The position of the unmanned water-spraying vehicle **20** is detected using a global navigation satellite system (GNSS). The position sensor **25A** includes a GNSS receiver and detects the position of the unmanned water-spraying vehicle **20** in the global coordinate system. The orientation sensor **25B** detects an orientation of the unmanned water-spraying vehicle **20**. A gyro sensor is exemplified as the orientation sensor **25B**. The speed sensor **25C** detects a traveling speed of the unmanned water-spraying vehicle **20**. As the speed sensor **25C**, a pulse sensor that detects the rotation of the wheel **26** is exemplified. The obstacle sensor **25D** detects an obstacle around the unmanned water-spraying vehicle **20**. The obstacle sensor **25D** detects an obstacle in a non-contact manner. Examples of the obstacle sensor **25D** include a laser sensor (light detection and ranging (LIDAR)) and a radar sensor (radio detection and ranging (RADAR)).

(33) The watering spray **28** sprays water in the tank **24**. The watering spray **28** is disposed at a rear of the tank **24**. The watering spray **28** sprays water behind the unmanned water-spraying vehicle **20**. In the embodiment, a plurality of the watering sprays **28** is provided. The plurality of watering sprays **28** are arranged at intervals in a vehicle width direction of the unmanned water-spraying vehicle **20** at the rear of the tank **24**. The vehicle width direction refers to a direction parallel to a rotation axis of the wheels **26** when the unmanned water-spraying vehicle **20** is in a straight traveling state.

(34) [Work Site]

(35) FIG. **4** is a schematic diagram illustrating a work site according to the embodiment. Examples of the work site include a mine or a quarry. The mine refers to a place or a place of business where minerals are mined. The quarry refers to a place or a place of business where stones are mined. At the work site, each of the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** operates.

(36) In the embodiment, the work site is a mine. Examples of the mine include a metal mine for mining metal, a non-metal mine for mining limestone, and a coal mine for mining coal.

(37) The work site is provided with a loading area **31**, an unloading area **32**, a parking area **33**, a refueling area **34**, a water supply area **35**, a travel path **36**, an intersection **37**, and a side path **38**.

(38) The loading area **31** is an area in which a loading operation of loading a load on the unmanned haul vehicle **10** is performed. In the loading area **31**, a loader **5** operates. As the loader **5**, an

excavator is exemplified.

(39) The unloading area **32** refers to an area where discharging work for discharging a load from the unmanned haul vehicle **10** is performed. A crusher **6** is provided in the unloading area **32**.

(40) The parking area **33** is an area where at least one of the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** is parked.

(41) The refueling area **34** is an area where at least one of the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** is supplied with fuel. An oil feeder **7** that supplies fuel is provided in the refueling area **34**.

(42) The water supply area **35** is an area where the unmanned water-spraying vehicle **20** is supplied with water. In the water supply area **35**, water for water-spraying is supplied to the tank **24**. The water supply area **35** is provided with a water supply device **8** that supplies water to the tank **24**.

(43) The travel path **36** refers to an area where an unmanned vehicle travels toward at least one of the loading area **31**, the unloading area **32**, the parking area **33**, the refueling area **34**, and the water supply area **35**. The travel path **36** is provided so as to connect at least the loading area **31** and the unloading area **32**. In the embodiment, the travel path **36** is connected to each of the loading area **31**, the unloading area **32**, the parking area **33**, the refueling area **34**, and the water supply area **35**.

(44) The intersection **37** refers to an area where a plurality of the travel paths **36** intersect or an area where one travel path **36** branches into a plurality of the travel paths **36**.

(45) The side path **38** is an area provided so as to branch from the travel path **36**. In the embodiment, the side path **38** is provided in the vicinity of the travel path **36**. A plurality of the side paths **38** are provided at intervals in the vicinity of the travel path **36**. The unmanned water-spraying vehicle **20** traveling on the travel path **36** can enter the side path **38** from the travel path **36**. The unmanned water-spraying vehicle **20** existing on the side path **38** can enter the travel path **36** from the side path **38**.

(46) [Management System]

(47) FIG. 5 is a functional block diagram illustrating the management system **1** of the work site according to the embodiment. The management system **1** includes a management device **2**, a communication system **3**, a control device **11**, and a control device **21**.

(48) The management device **2** includes a computer system. The management device **2** is connected to an input device **9**. The management device **2** includes a communication interface **41**, a storage circuit **42**, and a processing circuit **43**.

(49) The input device **9** is connected to the processing circuit **43**. The input device **9** is operated by an administrator of the control facility **4**. The input device **9** generates input data on the basis of an operation of the administrator. The input data generated by the input device **9** is input to the processing circuit **43**. Examples of the input device **9** include a touch panel, a computer keyboard, a mouse, and an operation button. Note that the input device **9** may be a non-contact type input device including an optical sensor, or may be a voice input device.

(50) The communication interface **41** is connected to the processing circuit **43**. The communication interface **41** controls communication between the management device **2** and at least one of the control device **11** and the control device **21**. The communication interface **41** communicates with at least one of the control device **11** and the control device **21** via the communication system **3**.

(51) The storage circuit **42** is connected to the processing circuit **43**. The storage circuit **42** stores data. As the storage circuit **42**, a nonvolatile memory or a volatile memory is exemplified. Examples of the nonvolatile memory include a read only memory (ROM) and a storage. Examples of the storage include a hard disk drive (HDD) and a solid state drive (SSD). As the volatile memory, a random access memory (RAM) is exemplified.

(52) The processing circuit **43** performs arithmetic processing and output processing of a control command. A processor is exemplified as the processing circuit **43**. Examples of the processor include a central processing unit (CPU) and a micro processing unit (MPU). A computer program is stored in the storage circuit **42**. The processing circuit **43** exerts a predetermined function by

acquiring and executing a computer program from the storage circuit **42**.

(53) The processing circuit **43** includes a first course data generation unit **61**, a second course data generation unit **62**, a traffic congestion prediction unit **63**, a guidance command unit **64**, a water-spraying data generation unit **65**, a first output unit **66**, and a second output unit **67**.

(54) The first course data generation unit **61** generates first course data indicating a travel condition of the unmanned haul vehicle **10** set at the work site. The first course data generation unit **61** may generate the first course data on the basis of the input data from the input device **9**.

(55) FIG. **6** is a diagram for explaining the first course data according to the embodiment. The first course data defines a travel condition of the unmanned haul vehicle **10**. The first course data includes a course point **101**, a travel course **102**, a target position of the unmanned haul vehicle **10**, a target orientation of the unmanned haul vehicle **10**, and a target traveling speed of the unmanned haul vehicle **10**.

(56) A plurality of the course points **101** are set on the travel path **36** including the intersection **37**. Furthermore, the plurality of course points **101** are set in each of the loading area **31**, the unloading area **32**, the parking area **33**, and the refueling area **34**. Each of the course points **101** defines a target position of the unmanned haul vehicle **10**. A target orientation and a target traveling speed of the unmanned haul vehicle **10** are set for each of the plurality of course points **101**. The plurality of course points **101** are set at intervals. The interval between the course points **101** is set to, for example, 1 [m] or more and 5 [m] or less. The intervals between the course points **101** may be uniform or non-uniform.

(57) The travel course **102** refers to a virtual line indicating a target travel route of the unmanned haul vehicle **10**. The travel course **102** is defined by a trajectory passing through the plurality of course points **101**. The unmanned haul vehicle **10** travels through the work site along the travel course **102**.

(58) The target position of the unmanned haul vehicle **10** refers to a target position of the unmanned haul vehicle **10** when passing through the course point **101**. The target position of the unmanned haul vehicle **10** may be defined in the local coordinate system of the unmanned haul vehicle **10** or may be defined in the global coordinate system.

(59) The target orientation of the unmanned haul vehicle **10** refers to a target orientation of the unmanned haul vehicle **10** when passing through the course point **101**.

(60) The target traveling speed of the unmanned haul vehicle **10** refers to a target traveling speed of the unmanned haul vehicle **10** when passing through the course point **101**. The target traveling speed of the unmanned haul vehicle **10** includes an upper limit speed (limit speed) indicating an upper limit value of the traveling speed of the unmanned haul vehicle **10**. The unmanned haul vehicle **10** travels at a traveling speed not exceeding the upper limit speed at the work site.

(61) The second course data generation unit **62** generates second course data indicating a travel condition of the unmanned water-spraying vehicle **20** set at the work site. The second course data generation unit **62** may generate the second course data on the basis of the input data from the input device **9**.

(62) FIG. **7** is a diagram for explaining the second course data according to the embodiment. The second course data defines a travel condition of the unmanned water-spraying vehicle **20**. The second course data includes a course point **201**, a travel course **202**, a target position of the unmanned water-spraying vehicle **20**, a target orientation of the unmanned water-spraying vehicle **20**, and a target traveling speed of the unmanned water-spraying vehicle **20**. A plurality of the course points **201** are set on the travel path **36**. Furthermore, the plurality of course points **201** are set in each of the loading area **31**, the unloading area **32**, the parking area **33**, the refueling area **34**, the water supply area **35**, and the side path **38**. The travel course **202** of the unmanned water-spraying vehicle **20** refers to a virtual line indicating a target travel route of the unmanned water-spraying vehicle **20**. The target traveling speed of the unmanned water-spraying vehicle **20** refers to a target traveling speed of the unmanned water-spraying vehicle **20** when passing through the

course point **201**. The target traveling speed of the unmanned water-spraying vehicle **20** includes an upper limit speed (speed limit) indicating an upper limit value of the traveling speed of the unmanned water-spraying vehicle **20**. The unmanned water-spraying vehicle **20** travels at a traveling speed not exceeding the upper limit speed at the work site. A function of the first course data and a function of the second course data are similar. Description of the second course data will be omitted.

(63) Note that the target traveling speed (upper limit speed) of the unmanned haul vehicle **10** and the target traveling speed (upper limit speed) of the unmanned water-spraying vehicle **20** are different. The target traveling speed (upper limit speed) of the unmanned water-spraying vehicle **20** is lower than the target traveling speed of the unmanned haul vehicle **10**.

(64) The traffic congestion prediction unit **63** predicts occurrence of a traffic congestion on the travel path **36** from travel situations of a plurality of the unmanned vehicles traveling on the travel path **36** at the work site.

(65) For example, the traffic congestion prediction unit **63** predicts that a traffic congestion will occur due to the unmanned water-spraying vehicle **20** when it is predicted that a distance between the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** will be less than or equal to a predetermined distance threshold after a predetermined time based on a travel situation of each of the unmanned haul vehicle **10** traveling on the travel path **36** and the unmanned water-spraying vehicle **20** traveling ahead of the unmanned haul vehicle **10**.

(66) When a predetermined first condition is satisfied, the guidance command unit **64** outputs a first guidance command so that the unmanned water-spraying vehicle **20** traveling on the travel path **36** enters the side path **38** from the travel path **36**. For example, when occurrence of a traffic congestion is predicted by the traffic congestion prediction unit **63**, the guidance command unit **64** outputs a first guidance command so that an unmanned vehicle that causes the traffic congestion enters the side path **38** branching from the travel path **36**. When occurrence of a traffic congestion caused by the unmanned water-spraying vehicle **20** is predicted, the guidance command unit **64** outputs a first guidance command so that the unmanned water-spraying vehicle **20** causing the traffic congestion enters the side path **38** from the travel path **36**. When a predetermined second condition is satisfied, the guidance command unit **64** outputs a second guidance command so that the unmanned water-spraying vehicle **20** existing on the side path **38** enters the travel path **36** from the side path **38**.

(67) The water-spraying data generation unit **65** generates water-spraying data for controlling the watering spray **28**. The water-spraying data includes at least one of execution and stop of water-spraying from the watering spray **28**, a water-spraying position where the watering spray **28** sprays water at the work site, and a water-spraying amount per unit time from the watering spray **28**. The water-spraying position where the watering spray **28** sprays water includes a water-spraying area where the watering spray **28** sprays water. Furthermore, in a case where a plurality of the watering sprays **28** is provided in the unmanned water-spraying vehicle **20**, the water-spraying data includes the number of the watering sprays **28** that execute water spraying. Furthermore, in a case where the watering spray **28** is installed at each of a plurality of positions of the unmanned water-spraying vehicle **20**, the water-spraying data includes an installation position of the watering spray **28** that executes water spraying. The water-spraying data generation unit **65** may generate the water-spraying data based on the input data from the input device **9**.

(68) The first output unit **66** outputs the first course data generated by the first course data generation unit **61** to the unmanned haul vehicle **10**. The first output unit **66** transmits the first course data from the communication interface **41** to the control device **11** of the unmanned haul vehicle **10**.

(69) The second output unit **67** outputs the second course data generated by the second course data generation unit **62** to the unmanned water-spraying vehicle **20**. The second output unit **67** transmits the second course data from the communication interface **41** to the control device **21** of the

unmanned water-spraying vehicle **20**.

(70) The second output unit **67** outputs at least one of the first guidance command and the second guidance command output from the guidance command unit **64** to the unmanned water-spraying vehicle **20**. The second output unit **67** transmits at least one of the first guidance command and the second guidance command from the communication interface **41** to the control device **21** of the unmanned water-spraying vehicle **20**.

(71) The second output unit **67** outputs the water-spraying data generated by the water-spraying data generation unit **65** to the unmanned water-spraying vehicle **20**. The second output unit **67** transmits the water-spraying data from the communication interface **41** to the control device **21** of the unmanned water-spraying vehicle **20**.

(72) The control device **11** includes a computer system. Similarly to the management device **2**, the control device **11** includes a communication interface, a storage circuit, and a processing circuit. The control device **11** includes a travel control unit **71** that controls the traveling device **13**. The travel control unit **71** controls the traveling device **13** based on the first course data transmitted from the management device **2**.

(73) The control device **21** includes a computer system. Similarly to the management device **2**, the control device **21** includes a communication interface, a storage circuit, and a processing circuit. The control device **21** includes a travel control unit **81** that controls the traveling device **23** and a water-spraying control unit **82** that controls the watering spray **28**. The travel control unit **81** controls the traveling device **23** based on the second course data transmitted from the management device **2**. The travel control unit **81** controls the traveling device **23** based on at least one of the first guidance command and the second guidance command transmitted from the management device **2**. The water-spraying control unit **82** controls the watering spray **28** based on the water-spraying data transmitted from the management device **2**.

(74) The travel control unit **71** controls the traveling device **13** based on the first course data and detected data of the sensor system **15**. The travel control unit **71** controls the traveling device **13** such that the unmanned haul vehicle **10** travels along the travel course **102** based on the detected data of the position sensor **15A** and the detected data of the orientation sensor **15B**.

(75) That is, the travel control unit **71** controls the traveling device **13** so that a deviation between a detected position of the unmanned haul vehicle **10** detected by the position sensor **15A** when passing through the course point **101** and a target position of the unmanned haul vehicle **10** set at the course point **101** becomes small.

(76) Furthermore, the travel control unit **71** controls the traveling device **13** so that a deviation between a detected orientation of the unmanned haul vehicle **10** detected by the orientation sensor **15B** when passing through the course point **101** and a target orientation of the unmanned haul vehicle **10** set at the course point **101** becomes small.

(77) Furthermore, the travel control unit **71** controls the traveling device **13** so that the unmanned haul vehicle **10** travels at a target traveling speed based on detected data of the speed sensor **15C**. That is, the travel control unit **71** controls the traveling device **13** so that a deviation between a detected traveling speed of the unmanned haul vehicle **10** detected by the speed sensor **15C** when passing through the course point **101** and a target traveling speed of the unmanned haul vehicle **10** set at the course point **101** becomes small.

(78) As described above, the target traveling speed of the unmanned haul vehicle **10** includes the upper limit speed of the unmanned haul vehicle **10**. The travel control unit **71** controls the traveling device **13** so that the unmanned haul vehicle **10** travels at a traveling speed not exceeding the upper limit speed based on the detected data of the speed sensor **15C**. That is, the travel control unit **71** controls the traveling device **13** so that the detected traveling speed of the unmanned haul vehicle **10** detected by the speed sensor **15C** when passing through the course point **101** does not exceed the upper limit speed of the unmanned haul vehicle **10** set at the course point **101**.

(79) The travel control unit **81** controls the traveling device **23** based on the second course data and

detected data of the sensor system **25**. The travel control unit **81** controls the traveling device **23** so that the unmanned water-spraying vehicle **20** travels along the travel course **202** based on detected data of the position sensor **25A** and detected data of the orientation sensor **25B**.

(80) That is, the travel control unit **81** controls the traveling device **23** so as to reduce a deviation between a detected position of the unmanned water-spraying vehicle **20** detected by the position sensor **25A** when passing through the course point **201** and a target position of the unmanned water-spraying vehicle **20** set at the course point **201**.

(81) Furthermore, the travel control unit **81** controls the traveling device **23** so as to reduce a deviation between a detected orientation of the unmanned water-spraying vehicle **20** detected by the orientation sensor **25B** when passing through the course point **201** and a target orientation of the unmanned water-spraying vehicle **20** set at the course point **201**.

(82) Furthermore, the travel control unit **81** controls the traveling device **23** based on detected data of the speed sensor **25C** so that the unmanned water-spraying vehicle **20** travels at a target traveling speed. That is, the travel control unit **81** controls the traveling device **23** so as to reduce a deviation between a detected traveling speed of the unmanned water-spraying vehicle **20** detected by the speed sensor **25C** when passing through the course point **101** and a target traveling speed of the unmanned water-spraying vehicle **20** set at the course point **201**.

(83) As described above, the target traveling speed of the unmanned water-spraying vehicle **20** includes the upper limit speed of the unmanned water-spraying vehicle **20**. The travel control unit **81** controls the traveling device **23** based on the detected data of the speed sensor **25C** so that the unmanned water-spraying vehicle **20** travels at a traveling speed not exceeding the upper limit speed. That is, the travel control unit **81** controls the traveling device **23** so that the detected traveling speed of the unmanned water-spraying vehicle **20** detected by the speed sensor **25C** when passing through the course point **201** does not exceed the upper limit speed of the unmanned water-spraying vehicle **20** set at the course point **201**.

(84) [Processing of Guidance Command Unit]

(85) Each of FIGS. **8**, **9**, and **10** is a diagram for explaining processing of the guidance command unit **64** according to the embodiment.

(86) The unmanned haul vehicle **10** travels on the travel path **36** at the work site at a first traveling speed **V1**. The first traveling speed **V1** is an actual traveling speed of the unmanned haul vehicle **10**. The first course data generation unit **61** generates first course data so that the unmanned haul vehicle **10** travels on the travel path **36** at the target traveling speed. The unmanned haul vehicle **10** travels on the travel path **36** at the first traveling speed **V1** based on the first course data transmitted from the first course data generation unit **61**.

(87) The unmanned water-spraying vehicle **20** travels on the travel path **36** at the work site at a second traveling speed **V2**. The second traveling speed **V2** is an actual traveling speed of the unmanned water-spraying vehicle **20**. The second course data generation unit **62** generates second course data so that the unmanned water-spraying vehicle **20** travels on the travel path **36** at the target traveling speed. The unmanned water-spraying vehicle **20** travels on the travel path **36** at the second traveling speed **V2** based on the second course data transmitted from the second course data generation unit **62**.

(88) Furthermore, the unmanned water-spraying vehicle **20** travels on the travel path **36** at the second traveling speed **V2** while spraying water on the travel path **36** at the work site. The water-spraying data generation unit **65** generates water-spraying data so that the unmanned water-spraying vehicle **20** travels while spraying water on the travel path **36**. The unmanned water-spraying vehicle **20** travels on the travel path **36** at the second traveling speed **V2** while spraying water on the travel path **36** based on the water-spraying data transmitted from the water-spraying data generation unit **65**. Note that the unmanned water-spraying vehicle **20** may travel on the travel path **36** at the second traveling speed **V2** without spraying water.

(89) Note that the first traveling speed **V1**, which is the actual traveling speed of the unmanned

haul vehicle **10**, and the target traveling speed of the unmanned haul vehicle **10** may not necessarily coincide with each other. For example, when the unmanned haul vehicle **10** travels on an uphill, the actual traveling speed of the unmanned haul vehicle **10** may be lower than the target traveling speed. Similarly, the second traveling speed **V2**, which is the actual traveling speed of the unmanned water-spraying vehicle **20**, and the target traveling speed of the unmanned water-spraying vehicle **20** may not necessarily coincide with each other.

(90) When a predetermined first condition is satisfied, the guidance command unit **64** outputs a first guidance command so that the unmanned water-spraying vehicle **20** traveling on the travel path **36** enters the side path **38** from the travel path **36**.

(91) FIG. **8** illustrates a state where the first condition is satisfied. The first condition includes a case where occurrence of a traffic congestion is predicted from the travel situations of the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** traveling on the travel path **36** at the work site.

(92) Traveling of the unmanned water-spraying vehicle **20** in front of the unmanned haul vehicle **10** includes setting the travel course **102** of the unmanned haul vehicle **10** and the travel course **202** of the unmanned water-spraying vehicle **20** on the same travel path **36**. Furthermore, the case where the unmanned water-spraying vehicle **20** travels in front of the unmanned haul vehicle **10** includes a case where the travel course **102** and the travel course **202** are set to be parallel to each other or the travel course **102** and the travel course **202** are set to overlap each other on the same travel path **36**.

(93) The case where occurrence of a traffic congestion is predicted includes a case where it is predicted that a distance between the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** will be less than or equal to a predetermined distance threshold after a predetermined time based on a travel situation of each of the unmanned haul vehicle **10** traveling on the travel path **36** and the unmanned water-spraying vehicle **20** traveling in front of the unmanned haul vehicle **10**.

(94) The guidance command unit **64** outputs a first guidance command to the unmanned water-spraying vehicle **20** so that the unmanned water-spraying vehicle **20** enters the side path **38** branching from the travel path **36** when occurrence of a traffic congestion is predicted from a travel situation of the unmanned haul vehicle **10** traveling on the travel path **36** at the work site at the first traveling speed **V1** and a travel situation of the unmanned water-spraying vehicle **20** traveling ahead of the unmanned haul vehicle **10** at the second traveling speed **V2** lower than the first traveling speed **V1**.

(95) FIG. **9** illustrates a state where the unmanned water-spraying vehicle **20** has entered the side path **38**. The second course data generation unit **62** generates second course data so that the second course data (travel course **202**) of the unmanned water-spraying vehicle **20** is set on the side path **38**. The travel course **202** of the unmanned water-spraying vehicle **20** is set on each of the travel path **36** and the side path **38**.

(96) The second course data generation unit **62** generates second course data such that the unmanned water-spraying vehicle **20** travels on the side path **38** at a traveling speed lower than the first traveling speed **V1**. On the basis of the second course data transmitted from the second course data generation unit **62**, the unmanned water-spraying vehicle **20** travels on the side path **38** at the second traveling speed **V2** or a traveling speed lower than the first traveling speed **V1**.

(97) Note that the second course data generation unit **62** may generate the second course data such that the unmanned water-spraying vehicle **20** stops on the side path **38**. The unmanned water-spraying vehicle **20** may stop on the side path **38** based on the second course data transmitted from the second course data generation unit **62**.

(98) In a case where the unmanned water-spraying vehicle **20** travels on the side path **38** at the second traveling speed **V2** or a traveling speed lower than the first traveling speed **V1**, the side path **38** functions as a low-speed traveling lane. In a case where the unmanned water-spraying vehicle

20 stops on the side path **38**, the side path **38** functions as a turnout.

(99) The unmanned haul vehicle **10** travels on the travel path **36** at the first traveling speed **V1** based on the first course data. The first course data is not set on the side path **38**. The unmanned haul vehicle **10** travels on the travel path **36** at the first traveling speed **V1** without entering the side path **38**.

(100) Since the unmanned haul vehicle **10** travels on the travel path **36** at a high speed in a state where the unmanned water-spraying vehicle **20** travels at a low speed or stops on the side path **38**, the unmanned haul vehicle **10** can overtake the unmanned water-spraying vehicle **20**.

(101) When a predetermined second condition is satisfied, the guidance command unit **64** outputs a second guidance command so that the unmanned water-spraying vehicle **20** existing on the side path **38** enters the travel path **36** from the side path **38**.

(102) FIG. **10** illustrates a state where the second condition is satisfied. The second condition includes that the unmanned haul vehicle **10** has overtaken the unmanned water-spraying vehicle **20** existing on the side path **38**.

(103) The situation in which the unmanned haul vehicle **10** has overtaken the unmanned water-spraying vehicle **20** includes that, when the unmanned haul vehicle **10** travels along the travel course **102** set on the travel path **36** and travels along the travel course **202** set on the travel path **36** after the unmanned water-spraying vehicle **20** enters the travel path **36** from the side path **38**, the unmanned haul vehicle **10** travels at the first traveling speed **V1** in front of the unmanned water-spraying vehicle **20** traveling at the second traveling speed **V2**.

(104) The guidance command unit **64** can determine whether or not the unmanned haul vehicle **10** has overtaken the unmanned water-spraying vehicle **20** based on the detected data of the position sensor **15A** of the unmanned haul vehicle **10** and the detected data of the position sensor **25A** of the unmanned water-spraying vehicle **20**. That is, the guidance command unit **64** determines whether or not the unmanned haul vehicle **10** has overtaken the unmanned water-spraying vehicle **20** based on a relative position between the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** at the work site.

(105) Traveling of the unmanned haul vehicle **10** in front of the unmanned water-spraying vehicle **20** includes setting the travel course **102** of the unmanned haul vehicle **10** and the travel course **202** of the unmanned water-spraying vehicle **20** on the same travel path **36**. Furthermore, the case where the unmanned haul vehicle **10** travels in front of the unmanned water-spraying vehicle **20** includes a case where the travel course **102** and the travel course **202** are set in parallel, or a case where the travel course **102** and the travel course **202** overlap each other on the same travel path **36**.

(106) The guidance command unit **64** outputs a second guidance command so that the unmanned water-spraying vehicle **20** enters the travel path **36** from the side path **38** after the unmanned haul vehicle **10** has overtaken the unmanned water-spraying vehicle **20** existing on the side path **38**. The second course data generation unit **62** generates second course data for allowing the unmanned water-spraying vehicle **20** to enter the travel path **36** from the side path **38**.

(107) The unmanned water-spraying vehicle **20** receives each of the second course data and the second guidance command from the management device **2**. After receiving the second guidance command, the unmanned water-spraying vehicle **20** enters the travel path **36** from the side path **38** based on the second course data.

(108) Since the unmanned haul vehicle **10** traveling at a high speed can overtake the unmanned water-spraying vehicle **20** traveling at a low speed, occurrence of a traffic congestion on the travel path **36** at the work site is suppressed.

(109) [Unmanned Vehicle Management Method]

(110) FIG. **11** is a flowchart illustrating an unmanned vehicle management method according to the embodiment.

(111) The first course data generation unit **61** generates first course data of the unmanned haul

vehicle **10**. The first output unit **66** transmits the first course data generated by the first course data generation unit **61** to the unmanned haul vehicle **10**. The unmanned haul vehicle **10** travels on the travel path **36** at the first traveling speed $V1$ based on the first course data (step **S1**).

(112) The second course data generation unit **62** generates second course data of the unmanned water-spraying vehicle **20**. The second output unit **67** transmits the second course data generated by the second course data generation unit **62** to the unmanned water-spraying vehicle **20**. Furthermore, in a case where the water-spraying data is generated by the water-spraying data generation unit **65**, the second output unit **67** transmits the water-spraying data to the unmanned water-spraying vehicle **20**. The unmanned water-spraying vehicle **20** travels on the travel path **36** at the second traveling speed $V2$ based on the second course data. Furthermore, when receiving the water-spraying data, the unmanned water-spraying vehicle **20** sprays water on the travel path **36** while traveling on the travel path **36** based on the water-spraying data (step **S2**).

(113) The traffic congestion prediction unit **63** predicts occurrence of a traffic congestion from a travel situation of each of the unmanned haul vehicle **10** traveling on the travel path **36** at the work site and the unmanned water-spraying vehicle **20** traveling in front of the unmanned haul vehicle **10** (step **S3**).

(114) For example, when it is predicted that a distance between the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** will be less than or equal to a predetermined distance threshold after a predetermined time based on a travel situation of each of the unmanned haul vehicle **10** traveling on the travel path **36** at the work site and the unmanned water-spraying vehicle **20** traveling ahead of the unmanned haul vehicle **10**, the traffic congestion prediction unit **63** predicts that a traffic congestion will occur due to the unmanned water-spraying vehicle **20**.

(115) The guidance command unit **64** determines whether or not the first condition is satisfied. That is, the guidance command unit **64** determines whether or not the unmanned water-spraying vehicle **20** travels in front of the unmanned haul vehicle **10** on the travel path **36**, and a distance after a predetermined time between the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** on the travel path **36** is less than or equal to a predetermined distance threshold, and determines whether or not it is predicted that a traffic congestion will occur (step **S4**).

(116) In a case where it is determined in step **S4** that the first condition is not satisfied (step **S4**: No), the process returns to steps **S1** and **S2**.

(117) In a case where it is determined in step **S4** that the first condition is satisfied (step **S4**: Yes), the guidance command unit **64** outputs a first guidance command to the unmanned water-spraying vehicle **20** so that the unmanned water-spraying vehicle **20** enters the side path **38** from the travel path **36** (step **S5**).

(118) The unmanned water-spraying vehicle **20** enters the side path **38** from the travel path **36** based on the first guidance command. The unmanned water-spraying vehicle **20** travels on the side path **38** at the second traveling speed $V2$ or a traveling speed lower than the first traveling speed lower than the second traveling speed $V2$. Note that the unmanned water-spraying vehicle **20** may stop on the side path **38** (step **S6**).

(119) The unmanned haul vehicle **10** travels on the travel path **36** at the first traveling speed $V1$ without entering the side path **38**.

(120) The guidance command unit **64** determines whether or not the second condition is satisfied. That is, the guidance command unit **64** determines whether or not the unmanned haul vehicle **10** has overtaken the unmanned water-spraying vehicle **20** existing on the side path **38** (step **S7**).

(121) In a case where it is determined in step **S7** that the second condition is not satisfied (step **S7**: No), the process returns to step **S6**.

(122) In a case where it is determined in step **S7** that the second condition is satisfied (step **S7**: Yes), the guidance command unit **64** outputs a second guidance command to the unmanned water-spraying vehicle **20** so that the unmanned water-spraying vehicle **20** enters the travel path **36** from the side path **38** (step **S8**).

(123) The unmanned water-spraying vehicle **20** enters the travel path **36** from the side path **38** based on the second guidance command. On the travel path **36**, the unmanned water-spraying vehicle **20** travels behind the unmanned haul vehicle **10** (step S9).

Advantageous Effects

(124) As described above, according to the embodiment, the occurrence of a traffic congestion is predicted from the travel situations of the plurality of unmanned vehicles traveling on the travel path **36** at the work site. In a case where it is predicted that the cause of the traffic congestion is the unmanned water-spraying vehicle **20**, the first guidance command is output so that the unmanned water-spraying vehicle **20** that causes the traffic congestion enters the side path **38** branching from the travel path **36**. As a result, the unmanned haul vehicle **10** can overtake the unmanned water-spraying vehicle **20** by continuing traveling on the travel path **36**. Therefore, the occurrence of a traffic congestion on the travel path **36** at the work site is suppressed.

(125) When it is predicted that the unmanned haul vehicle **10** traveling on the travel path **36** at the first traveling speed **V1** at the work site approaches the unmanned water-spraying vehicle **20** traveling in front of the unmanned haul vehicle **10**, a first guidance command is output to the unmanned water-spraying vehicle **20** so that the unmanned water-spraying vehicle **20** enters the side path **38** branching from the travel path **36**. As a result, the unmanned haul vehicle **10** can overtake the unmanned water-spraying vehicle **20** by continuing traveling on the travel path **36**. Therefore, the occurrence of a traffic congestion on the travel path **36** at the work site is suppressed.

(126) The first condition under which the first guidance command is output includes that a distance after a predetermined time between the unmanned haul vehicle **10** and the unmanned water-spraying vehicle **20** on the travel path **36** has become less than or equal to a predetermined distance threshold. As a result, the first guidance command is output when the first condition is appropriately determined and the unmanned water-spraying vehicle **20** needs to overtake the unmanned haul vehicle **10**.

(127) The second condition for outputting the second guidance command includes that the unmanned haul vehicle **10** traveling on the travel path **36** has overtaken the unmanned water-spraying vehicle **20** existing on the side path **38**. As a result, the second condition is appropriately determined, and the unmanned water-spraying vehicle **20** can travel behind the unmanned haul vehicle **10** on the travel path **36**.

Other Embodiments

(128) In the above-described embodiment, the first course data may not be generated. The unmanned haul vehicle **10** may travel on the travel path **36** at the first traveling speed **V1** not based on the first course data. In the above-described embodiment, the second course data may not be generated. The unmanned water-spraying vehicle **20** may travel on the travel path **36** at the second traveling speed **V2** not based on the second course data. The unmanned water-spraying vehicle **20** may travel on the side path **38** at the second traveling speed **V2** or a traveling speed lower than the first traveling speed **V1**, or may decelerate or stop on the side path **38** not based on the second course data.

(129) In the above-described embodiment, at least a part of the function of the control device **11** and the function of the control device **21** may be provided in the management device **2**, or at least a part of the function of the management device **2** may be provided in one or both of the control device **11** and the control device **21**. For example, in the above-described embodiment, the control device **11** may have the function of the first course data generation unit **61** and the function of the traffic congestion prediction unit **63**. The control device **21** may have the function of the second course data generation unit **62**, the function of the traffic congestion prediction unit **63**, the function of the guidance command unit **64**, and the function of the water-spraying data generation unit **65**.

(130) In the above-described embodiment, each of the first course data generation unit **61**, the second course data generation unit **62**, the traffic congestion prediction unit **63**, the guidance

command unit **64**, the water-spraying data generation unit **65**, the first output unit **66**, and the second output unit **67** may be configured by different hardware.

(131) In the above-described embodiment, the first unmanned vehicle **10** is an unmanned haul vehicle, and the second unmanned vehicle **20** is an unmanned water-spraying vehicle. Both the first unmanned vehicle **10** and the second unmanned vehicle **20** may be unmanned haul vehicles. The first unmanned vehicle **10** may be an unmanned haul vehicle that travels on the travel path **36** at the first traveling speed **V1**, and the second unmanned vehicle **20** may be an unmanned haul vehicle that travels on the travel path **36** at the second traveling speed **V2**. Furthermore, both the first unmanned vehicle **10** and the second unmanned vehicle **20** may be unmanned water-spraying vehicles. The first unmanned vehicle **10** may be an unmanned water-spraying vehicle that travels on the travel path **36** at the first traveling speed **V1**, and the second unmanned vehicle **20** may be an unmanned water-spraying vehicle that travels on the travel path **36** at the second traveling speed **V2**.

REFERENCE SIGNS LIST

(132) **1** MANAGEMENT SYSTEM **2** MANAGEMENT DEVICE **3** COMMUNICATION SYSTEM **3A** WIRELESS COMMUNICATION DEVICE **3B** WIRELESS COMMUNICATION DEVICE **3C** WIRELESS COMMUNICATION DEVICE **4** CONTROL FACILITY **5** LOADER **6** CRUSHER **7** OIL FEEDER **8** WATER SUPPLY DEVICE **9** INPUT DEVICE **10** UNMANNED HAUL VEHICLE (FIRST UNMANNED VEHICLE) **11** CONTROL DEVICE **12** VEHICLE BODY **13** TRAVELING DEVICE **14** DUMP BODY **15** SENSOR SYSTEM **15A** POSITION SENSOR **15B** ORIENTATION SENSOR **15C** SPEED SENSOR **15D** OBSTACLE SENSOR **16** WHEEL **16F** FRONT WHEEL **16R** REAR WHEEL **17** TIRE **17F** FRONT TIRE **17R** REAR TIRE **20** UNMANNED WATER-SPRAYING VEHICLE (SECOND UNMANNED VEHICLE) **21** CONTROL DEVICE **22** VEHICLE BODY **23** TRAVELING DEVICE **24** TANK **25** SENSOR SYSTEM **25A** POSITION SENSOR **25B** ORIENTATION SENSOR **25C** SPEED SENSOR **25D** OBSTACLE SENSOR **26** WHEEL **26F** FRONT WHEEL **26R** REAR WHEEL **27** TIRE **27F** FRONT TIRE **27R** REAR TIRE **28** WATERING SPRAY **29** CAB **31** LOADING AREA **32** UNLOADING AREA **33** PARKING AREA **34** REFUELING AREA **35** WATER SUPPLY AREA **36** TRAVEL PATH **37** INTERSECTION **38** SIDE PATH **41** COMMUNICATION INTERFACE **42** STORAGE CIRCUIT **43** PROCESSING CIRCUIT **61** FIRST COURSE DATA GENERATION UNIT **62** SECOND COURSE DATA GENERATION UNIT **63** TRAFFIC CONGESTION PREDICTION UNIT **64** GUIDANCE COMMAND UNIT **65** WATER-SPRAYING DATA GENERATION UNIT **66** FIRST OUTPUT UNIT **67** SECOND OUTPUT UNIT **71** TRAVEL CONTROL UNIT **81** TRAVEL CONTROL UNIT **82** WATER-SPRAYING CONTROL UNIT **101** COURSE POINT **102** TRAVEL COURSE **201** COURSE POINT **202** TRAVEL COURSE

Claims

1. An unmanned vehicle management system comprising: a traffic congestion prediction unit that predicts occurrence of a traffic congestion from travel situations of a plurality of unmanned vehicles traveling on a travel path at a work site, the plurality of unmanned vehicles including a first unmanned vehicle and a second unmanned vehicle; and a guidance command unit that, when occurrence of a traffic congestion is predicted, outputs a first guidance command such that the second unmanned vehicle that causes the traffic congestion enters a side path branching from the travel path; wherein a first course data generation unit generates first course data indicating a travel condition of the first unmanned vehicle, wherein a second course data generation unit generates second course data indicating a travel condition of the second unmanned vehicle, and wherein the second course data includes a travel course indicating a target travel route of the second unmanned vehicle, and the travel course is set on each of the travel path and the side path.
2. The unmanned vehicle management system according to claim 1, wherein based on a travel

- situation of each of the first unmanned vehicle traveling on the travel path and the second unmanned vehicle traveling ahead of the first unmanned vehicle, the traffic congestion prediction unit predicts that a traffic congestion will occur due to the second unmanned vehicle when a distance between the first unmanned vehicle and the second unmanned vehicle is predicted to be less than or equal to a predetermined distance threshold after a predetermined time.
3. The unmanned vehicle management system according to claim 2, wherein the first course data generation unit generates the first course data such that the first unmanned vehicle travels on the travel path at a first traveling speed.
 4. The unmanned vehicle management system according to claim 3, wherein the second course data generation unit generates the second course data such that the second unmanned vehicle travels on the travel path at a second traveling speed.
 5. The unmanned vehicle management system according to claim 4, wherein the second course data generation unit generates the second course data such that the second unmanned vehicle travels on the side path at the second traveling speed or a traveling speed lower than the first traveling speed.
 6. The unmanned vehicle management system according to claim 4, wherein the second course data generation unit generates the second course data such that the second unmanned vehicle stops on the side path.
 7. The unmanned vehicle management system according to claim 2, wherein the guidance command unit outputs a second guidance command such that the second unmanned vehicle enters the travel path from the side path after the first unmanned vehicle overtakes the second unmanned vehicle existing on the side path.
 8. The unmanned vehicle management system according to claim 2, wherein the first unmanned vehicle is an unmanned haul vehicle, and the second unmanned vehicle is an unmanned water-spraying vehicle.
 9. An unmanned vehicle management method comprising: predicting occurrence of a traffic congestion from travel situations of a plurality of unmanned vehicles traveling on a travel path at a work site, the plurality of unmanned vehicles including a first unmanned vehicle and a second unmanned vehicle; when occurrence of a traffic congestion is predicted, causing the second unmanned vehicle that causes the traffic congestion to enter a side path branching from the travel path; generating first course data indicating a travel condition of the first unmanned vehicle; and generating second course data indicating a travel condition of the second unmanned vehicle, wherein the second course data includes a travel course indicating a target travel route of the second unmanned vehicle, and the travel course is set on each of the travel path and the side path.
 10. The unmanned vehicle management method according to claim 9, wherein based on a travel situation of each of a first unmanned vehicle traveling on the travel path and a second unmanned vehicle traveling ahead of the first unmanned vehicle, it is predicted that a traffic congestion will occur due to the second unmanned vehicle when a distance between the first unmanned vehicle and the second unmanned vehicle is predicted to be less than or equal to a predetermined distance threshold after a predetermined time.
 11. The unmanned vehicle management method according to claim 10, wherein the first unmanned vehicle travels on the travel path at a first traveling speed based on first course data indicating a travel condition of the first unmanned vehicle, and the second unmanned vehicle travels on the travel path at a second traveling speed based on second course data indicating a travel condition of the second unmanned vehicle.
 12. The unmanned vehicle management method according to claim 11, wherein the second unmanned vehicle travels on the side path at the second traveling speed or a traveling speed lower than the first traveling speed.
 13. The unmanned vehicle management method according to claim 10, wherein the second unmanned vehicle stops on the side path.
 14. The unmanned vehicle management method according to claim 10, further comprising causing

the second unmanned vehicle to enter the travel path from the side path after the first unmanned vehicle overtakes the second unmanned vehicle existing on the side path.
